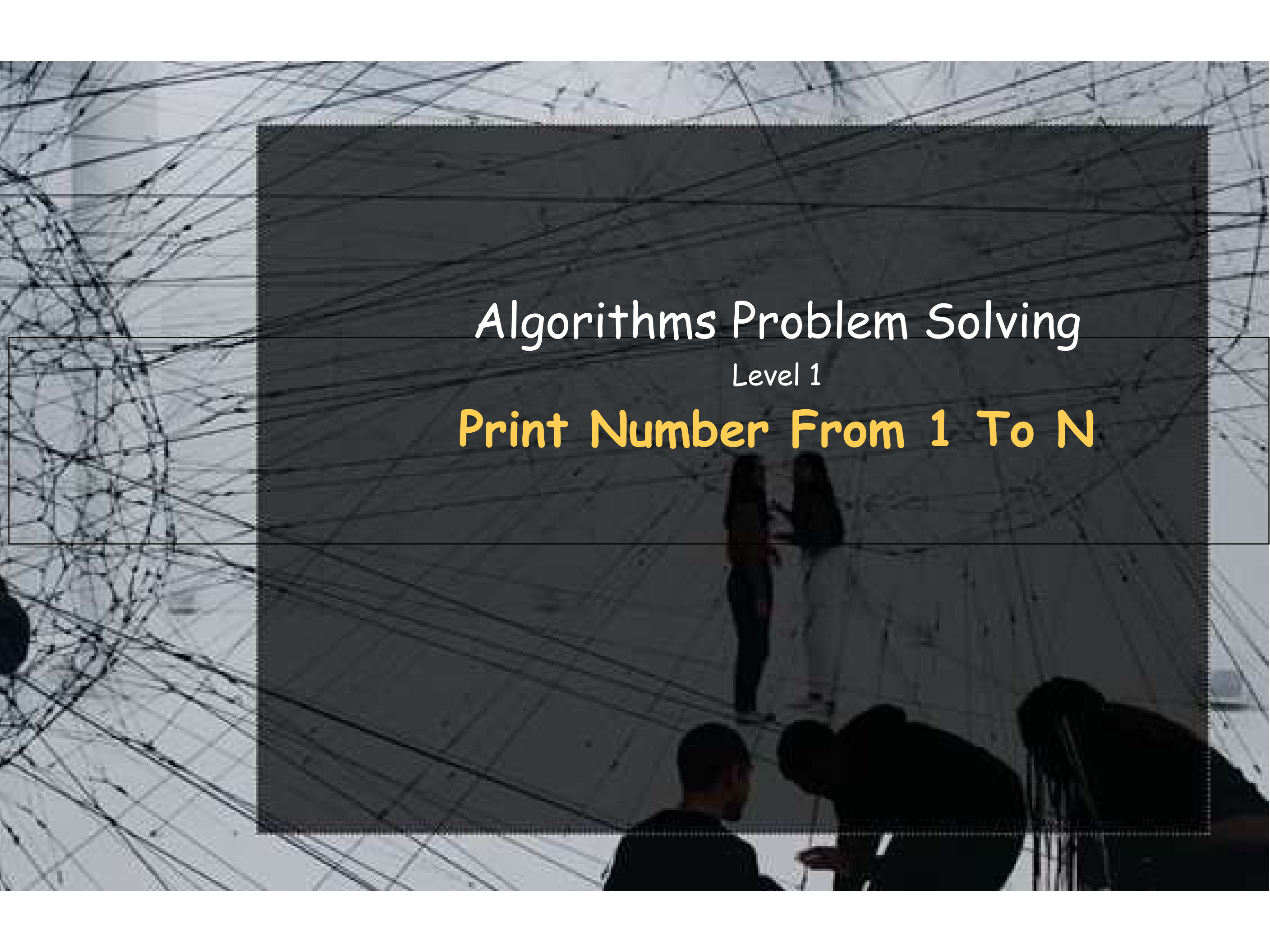


Algorithms Problem Solving

Level 1

Print Number From 1 To
N



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Print Number From 1 To N

Definition

- ◆ This program reads the user's age and checks whether it is between 18 and 45.
- ◆ If the age is within this range, the program prints "Eligible for job".
- ◆ Otherwise, it prints "Not eligible for job".

Explanation

- ◆ ``readNumb()`` asks the user to enter a number and returns it.
- ◆ ``printRangeFrom1ToN()`` receives N and prints the numbers from 1 through N.
- ◆ The ``for`` loop starts with ``counter = 1``.
- ◆ ``counter <= N`` means the loop continues as long as the counter is less than or equal to N.
- ◆ ``counter++`` increases the counter by 1 after each iteration.
- ◆ ``printRangeFrom1ToN(readNumb())`` first reads the number, then passes it directly to the printing function.

Steps in main()

1. Read the number N from the user.
2. Set `counter` to 1.
3. Check whether `counter` $\leq N$.
4. Print the current value of `counter`.
5. Increase `counter` by 1.
6. Repeat until `counter` becomes greater than N .

💡 Example

◆ If the user enters:

$N = 5$

◆ The loop starts at 1 and continues until 5.

➡ Result

Range:

1

2

3

4

5

Note

- ◆ ``counter <= N`` includes N itself in the output.
- ◆ If the condition were ``counter < N``, the program would stop at $N - 1$.

⊘ Warning

⚡ If the user enters a negative number, the loop will not execute because `counter` starts at 1 and `1 <= N` will already be false.



Comparison

▨ `counter <= N`

- ✓ Prints numbers from 1 through N.
- ✓ Includes N.
- ✓ Example: $N = 5 \rightarrow 1, 2, 3, 4, 5$.

▤ `counter < N`

- ✓ Stops before N.
- ✓ Does not include N.
- ✓ Example: $N = 5 \rightarrow 1, 2, 3, 4$.

✦ Summary

- ✓ The program prints numbers from 1 to N.
- ✓ The `for` loop controls the repetition.
- ✓ `counter` starts at 1 and increases by 1.
- ✓ `counter <= N` ensures that N is included.
- ✓ `readNumb()` and `printRangeFrom1ToN()` separate input from output.

Thank you 😊